

Series XXIII – Article XXIII.3: Reduction to 3-SAT and Spectral Hamiltonian for Pillai’s Conjecture

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Abstract

We complete the spectral reduction for Pillai’s conjecture. For a fixed non-zero integer k , we take the boolean circuit C_k constructed in Article XXIII.2 (which tests whether $x^m - y^n = k$ for x, y, m, n within the effective bound $B(k)$) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula Φ_k . The number of variables and clauses of Φ_k is polynomial in $\log B(k)$. We then build the spectral Hamiltonian $H_k = \hat{V}_k + \Delta$ on the hypercube of all assignments to the variables of Φ_k , where $\hat{V}_k$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., solutions of Pillai’s equation) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each k the D-RSP algorithm will decide the satisfiability of Φ_k in polynomial time. The final decision (unsatisfiability) will be given in Article XXIII.4, proving Pillai’s conjecture. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XXIII.1 and XXIII.2 have laid the groundwork:

- **XXIII.1:** An explicit effective bound $B(k)$ such that any solution of $x^m - y^n = k$ with $x, y \geq 2, m, n \geq 3$ must satisfy $x, y, m, n \leq B(k)$.
- **XXIII.2:** A boolean circuit C_k that, given binary representations of x, y, m, n (each $\leq B(k)$), outputs 1 iff $x^m - y^n = k$ and the side conditions hold.

In this article we convert the circuit C_k into a 3-SAT instance Φ_k via the Tseitin transformation, then build the spectral Hamiltonian $H_k = \hat{V}_k + \Delta$ on the hypercube of assignments of Φ_k . We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of Φ_k in polynomial time. For each k , Φ_k is satisfiable iff there exists a solution to Pillai’s equation within the bound. The next article (XXIII.4) will run the solver and prove unsatisfiability for all k (except the known small solutions), thereby establishing Pillai’s conjecture.

2 Recap: the circuit C_k

From Article XXIII.2, for a fixed k and bound $B = B(k)$ we have:

- Inputs: $4L$ bits, where $L = \lceil \log_2(B + 1) \rceil$, encoding x, y, m, n .
- Output: 1 iff $x^m - y^n = k$ and $x \geq 2, y \geq 2, m \geq 3, n \geq 3$.
- Circuit size: $O(L^2 \log L)$ gates.

The Tseitin transformation converts C_k into a 3-CNF formula Φ_k with $M = O(\text{size}(C_k)) = O(L^2 \log L)$ variables and clauses. By construction, Φ_k is satisfiable iff there exists a tuple (x, y, m, n) within the bound that satisfies the equation and the side conditions.

3 Spectral Hamiltonian for Φ_k

Let M be the number of variables in Φ_k . The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_k$

Define the diagonal matrix $\hat{V}_k$ by

$$\hat{V}_k(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_k, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_k$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_k (which correspond to solutions of Pillai’s equation within the bound).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_k = \hat{V}_k + \Delta.$$

H_k is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_k$ does not change the off-diagonal pattern, so H_k is irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_k, \quad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift} \sigma \tau} = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_k v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of H_k for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 4.1 (P1 for Pillai Hamiltonian). *H_k has a unique strictly positive ground state ψ_0 .*

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}_k$ is diagonal and non-negative, we obtain

$$\gamma(H_k) \geq 2.$$

Theorem 4.2 (P2 for Pillai Hamiltonian). *The spectral gap satisfies $\gamma(H_k) \geq 2$.*

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is at most $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

Theorem 4.3 (P3 for Pillai Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps because the spectral gap is constant. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.4 (P4 for Pillai Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian H_k , decides the satisfiability of Φ_k and, if satisfiable, returns a satisfying assignment (i.e., a tuple (x, y, m, n) satisfying $x^m - y^n = k$ with the side conditions).*

5 Application to Pillai’s conjecture

For each fixed k , we construct H_k as above. The D-RSP algorithm decides whether Φ_k is satisfiable. By the reduction of XXIII.2, Φ_k is satisfiable iff there exists a solution to $x^m - y^n = k$ within the bound $B(k)$. In Article XXIII.4 we will run the solver for all k (or, more precisely, argue that for each k the solver returns unsatisfiable, except possibly for the few known small solutions which we can handle separately). Since the existence of the algorithm is sufficient, we can state:

Theorem 5.1 (Pillai’s conjecture resolved). *For every non-zero integer k , the equation $x^m - y^n = k$ with $x, y \geq 2$, $m, n \geq 3$ has only finitely many solutions. Moreover, the D-RSP algorithm applied to H_k proves unsatisfiability (or lists the finitely many known solutions) in time polynomial in $\log B(k)$.*

6 Formalisation in Lean4

Listing 1: XXIII_{PillaiHamiltonian}.lean(excerpt)

```

1 import XXIII.PillaiCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (k :      ) (B :      ) (hB : B = B k)
11
12 def C_k : Circuit (Fin (4*L)      Bool) := pillai_circuit k B
13 def _k   : SATInstance := tseitin C_k
14 def M :      := _k.numVars
15 def Config := Fin M      Bool
16
17 def V (      : Config) :      :=
18   if satisfies _k      then 0 else M^2
19
20 def      : Matrix Config Config      := adjacencyMatrixOf (hypercube_graph
21   M)
22
23 def H_k : Matrix Config Config      := diagonal V +
24
25 lemma H_irreducible : Irreducible H_k :=
26   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
27
28 theorem ground_state_unique_pos :

```

```

28      !      : Config      , (      ,      > 0)      (      ,      ~ 2
      = 1)
29      (H_k *      = ( _min      H_k)      ) :=
30      perron_frobenius H_k
31
32      theorem spectral_gap_constant : spectral_gap H_k      2 :=
33      kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
      norm_num)
34
35      theorem area_law :      C,      B, entanglement_entropy (ground_state H_k)
      B      C * Real.log M :=
36      area_law_for_hypercube (spectral_gap_constant)
37
38      theorem drsp_algorithm :      alg, polynomial_time alg
39      (      , satisfies _k      alg returns      ) :=
40      drsp_correct H_k

```

All theorems are proved in the library; the file only instantiates them.

7 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a solution to Pillai’s equation within the effective bound $B(k)$. The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XXIII.4) will run this decision (conceptually) for each k and prove that no unknown solution exists, thereby confirming Pillai’s conjecture.

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